

Certificate C: Mu4-Equivariance of the Corrections

$22 = 1 + 7 + 7 + 7$ and the regular representation

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 11/16

1 Setting

Certificate C establishes the naturality of the corrections: the triple $(7, 22, \lambda_1)$ is consistent with the μ_4 representation — the group of quarter-turns present on all three stands.

Theorem 1 (C1 — K3 isotypy). *In H^2 of the Fermat quartic the isotypic decomposition over the μ_4 characters is $22 = 1 + 7 + 7 + 7$. Three independent counting routes (direct orbits, the spectrum of the permutation representation, exact character multiplicities) agree.*

Proof. The action of $(\mu_4)^2$ on the 48 lines has 12 orbits. The permutation representation decomposes over the characters; projection operators $\Pi_\chi = \frac{1}{4} \sum_k i^{-k} R^k$ extract the isotypic parts. The orbit count gives $(1, 7, 7, 7)$; the spectrum of the permutation representation gives the same multiplicities as eigenspace dimensions; the pairing with exact line characters is the third route. Three agreeing routes exclude coincidence. $\square$

Theorem 2 (C2 — torus, regular representation). *On the torus the correction operator λ_1 commutes with μ_4 ; the characteristic polynomial is $x^4 - 1$ — the regular representation; the multiplicity of λ_1 is 4; the continuum is $\lambda_1 = 4\pi^2$; the triple $(4, 1, 4\pi^2)$ is consistent.*

Proof. Commutativity — the equivariance of the Laplacian; the eigenspaces are invariant and decompose over the μ_4 characters. The polynomial $x^4 - 1$ has simple roots $\{1, -1, i, -i\}$ — each character exactly once: the regular representation. The value $4\pi^2$ is realized on every character — the lattice continuum. $\square$

Protocol data: the Klein heptad

Φ_7 — the cyclotomic polynomial; $\Delta_{49a1} = -343 = -7^3$; $j = -3375 = -15^3$; the CM order $\mathbb{Z}[(1 + \sqrt{-7})/2]$; $h(-7) = 1$; τ certified. The heptad $\sqrt{-7}$ is the exact integer environment of the phase $\pi/7$. The μ_4 stand: anomaly = i ; 42 elements of order 4 in the dimension-32 permutation representation; μ_4 on all three stands. Grid: symmetric violations 0; $(7, 22)$ violations 1023/1024.

Summary of results: summary

Certificate C accepted (21/21 checks): the isotypy $22 = 1 + 7 + 7 + 7$ confirmed by three routes; the torus gives the regular representation; the seven receives its exact arithmetic face.

Verification: reproduction

`python3 laboratory.py` → “Certificates → C”: projection operators, orbits, spectrum — all exact.